

Contexts and pseudocontexts

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Abstract

In the projection lattice of a finite-dimensional Hilbert space, an orthogonal basis—a *context*—is characterised by the property that its atomic probabilities sum to one on every state. We study a weakening of this condition in which two tuples of atoms induce the same probability sum on every state. Such pairs are called *pseudocontexts*. In finite dimension this is equivalent to two distinct rank-one decompositions of the same positive operator on its support, placing the notion naturally among finite frame-theoretic constructions. Size $k = 1$ is trivial; size $k = 2$ already admits genuinely nonorthogonal examples over both $\mathbb{R}$ and $\mathbb{C}$; and, in fact, examples exist for every $k \geq 2$. We present the operator formulation, give a general explicit construction, analyse the spectral structure of the associated structure operator, illustrate the symmetric $k = 3$ case, and list the remaining classification problems.

1 Introduction

Let L be the lattice of projections in a Hilbert space of finite dimension $n \geq 3$. Its atoms (minimal non-zero elements) correspond to one-dimensional subspaces, represented by nonzero vectors. States are described by probability measures $P: L \rightarrow [0, 1]$. We denote by $S(L)$ the set of all states on L .

Vectors $a_1, \dots, a_n$ form a *context* if they compose an orthogonal basis. In this case (and only in this case) they satisfy

$$\forall P \in S(L) : \sum_{i=1}^n P(a_i) = 1. \quad (1)$$

The present note concerns weaker additive identifications obtained by comparing two tuples of atoms that are not required to be orthogonal. Such *pseudocontexts* are quantum-logical shadows of orthomodular-lattice constructions of Mayet–Navara–Rogalewicz [3] and have direct physical meaning: they encode state-independent empirical equivalences between families of generally noncommuting observables. The answer is simple: pseudocontexts already exist for $k = 2$, and, more generally, for every $k \geq 2$.

2 Preliminaries

Throughout, H is a complex Hilbert space with $\dim H = n \geq 3$, and $L = L(H)$ is the orthomodular lattice of its closed subspaces. Its atoms correspond to one-dimensional subspaces, or equivalently to *rays* spanned by nonzero vectors. We identify an atom with

any representative unit vector a and write $P_a = |a\rangle\langle a|$ for the orthogonal projection onto the ray it spans.

A *state* is a map $P: L \rightarrow [0, 1]$ with $P(I) = 1$ that is finitely additive on orthogonal families. By Gleason's theorem [1], every state arises from a density operator ρ via

$$P(a) = \text{tr}(\rho P_a), \quad \rho \geq 0, \quad \text{tr} \rho = 1. \quad (2)$$

The key consequence we use repeatedly is that two self-adjoint operators that agree on every state are equal as operators.

3 Contexts and pseudocontexts

The operator form of (1) is elementary and is the starting point for everything that follows.

Proposition 1 (Contexts). *Unit vectors $a_1, \dots, a_n \in H$ form an orthonormal basis if and only if*

$$\sum_{i=1}^n P_{a_i} = I. \quad (3)$$

Proof. The forward implication is immediate. For the converse, fix j and evaluate (3) on the vector a_j :

$$1 = \langle a_j, I a_j \rangle = \sum_{i=1}^n |\langle a_i, a_j \rangle|^2.$$

The term with $i = j$ already contributes 1, so all others must vanish. Hence $\langle a_i, a_j \rangle = 0$ for $i \neq j$. As this holds for every j , the family is orthonormal, and being of size n in an n -dimensional space it is an orthonormal basis. $\square$

Combined with (2), Proposition 1 recovers (1) as a state-theoretic characterisation of orthogonal bases.

Definition 2 (Pseudocontext). Let $k \geq 1$. A *pseudocontext* of size k is a pair of k -tuples of atoms $(a_1, \dots, a_k), (b_1, \dots, b_k)$ such that

- (i) each tuple contains pairwise distinct atoms,
- (ii) the two tuples are disjoint as sets of rays, so that $|\{a_1, \dots, a_k, b_1, \dots, b_k\}| = 2k$, and
- (iii) $\forall P \in S(L) : \sum_{i=1}^k P(a_i) = \sum_{i=1}^k P(b_i)$.

A pseudocontext is called *genuine* if neither k -tuple is a context.

Remark 3. Condition (i) excludes repeated atoms within a single tuple; condition (ii) ensures the two tuples are truly distinct. Both are part of the definition; without them the notion degenerates trivially.

By (2), condition (iii) of Definition 2 is equivalent to the operator identity

$$T := \sum_{i=1}^k P_{a_i} = \sum_{i=1}^k P_{b_i}, \quad (4)$$

which we call the *structure operator* of the pseudocontext. Thus pseudocontexts are precisely pairs of distinct rank-one decompositions of a single positive operator on its support. This places them naturally among finite frame-theoretic constructions: contexts are the special case in which the structure operator is the identity on the relevant subspace; in general, pseudocontexts are rank-one decompositions of a common positive operator. If the common operator is proportional to the identity on its support, the family is a tight frame; after normalisation on that support one obtains a POVM.

Remark 4 (Connection to SIC-POVMs). When the common structure operator satisfies $T = \frac{k}{n}I$ (tight frame on all of H), each family $\{P_{a_i}\}$ and $\{P_{b_i}\}$ scaled by n/k forms a symmetric informationally complete POVM (SIC-POVM) if $k = n^2$ and the frame vectors are equiangular [9, 10]. Pseudocontexts therefore generalise the SIC-POVM notion by dropping both the tightness and equiangularity requirements.

4 The smallest cases

Proposition 5 (Size $k = 1$). *There is no pseudocontext of size $k = 1$. This holds for every $n \geq 1$.*

Proof. The condition $P(a_1) = P(b_1)$ for all $P \in S(L)$ is equivalent, via (2), to $P_{a_1} = P_{b_1}$. This rank-one identity forces a_1 and b_1 to span the same ray, contradicting distinctness condition (ii). $\square$

Remark 6. Every pseudocontext of size $k \geq 2$ can be realised in a two-dimensional subspace (and hence embedded into $\mathbb{C}^n$ for any $n \geq 2$): all constructions in Sections 4.1 and 5 live in $\text{span}\{e_1, e_2\}$. The case $\mathbb{C}^1$ is excluded since every unit vector spans the unique ray, making the distinctness condition $|\{a_i\} \cup \{b_i\}| = 2k$ impossible.

4.1 Pseudocontexts of size $k = 2$

Corollary 7 (Size $k = 2$, complex). *There are genuinely nonorthogonal pseudocontexts of size $k = 2$ in $\mathbb{C}^n$ for every $n \geq 2$.*

Proof. Embed $\mathbb{C}^2$ into $\mathbb{C}^n$ as the span of the first two basis vectors and set

$$u_{\pm} = \left(\frac{\sqrt{3}}{2}, \pm\frac{1}{2}, 0, \dots, 0\right), \quad v_{\pm} = \left(\frac{\sqrt{3}}{2}, \pm\frac{i}{2}, 0, \dots, 0\right).$$

Then

$$P_{u_+} + P_{u_-} = P_{v_+} + P_{v_-} = \begin{pmatrix} 3/2 & 0 \\ 0 & 1/2 \end{pmatrix} \oplus 0_{n-2}.$$

The four rays are distinct. Within each pair, $\langle u_+, u_- \rangle = -\frac{1}{4} \neq 0$ and $\langle v_+, v_- \rangle = \frac{1}{4} \neq 0$, so neither pair is orthogonal. Hence neither pair is a context, and the pseudocontext is genuine. $\square$

Remark 8 (Real pseudocontexts of size $k = 2$). Genuine pseudocontexts of size $k = 2$ exist over *real* Hilbert spaces as well. In $\mathbb{R}^2$ consider the operator $T = \text{diag}(3/2, 1/2)$ and the two decompositions

$$u_{\pm} = \left(\frac{\sqrt{3}}{2}, \pm\frac{1}{2}\right), \quad v_{\pm} = \left(\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right).$$

One checks directly that $P_{u_+} + P_{u_-} = P_{v_+} + P_{v_-} = T$, the four rays are distinct, and $\langle u_+, u_- \rangle = -\frac{1}{4} \neq 0$, so the pseudocontext is genuine. Thus complex phases are *not* necessary for $k = 2$; they merely provide one convenient route to the construction. For $k \geq 3$, real pseudocontexts also exist, as demonstrated by Example 11.

5 Pseudocontexts of every size $k \geq 2$

Theorem 9 (Complex-phase construction). *Let H contain an orthonormal pair e_1, e_2 . Fix an integer $k \geq 2$, let $\omega = e^{2\pi i/k}$, choose $\alpha \in (0, \pi/2)$, and choose $\phi \in \mathbb{R}$ such that $e^{i\phi}$ is not a k th root of unity. Define unit vectors*

$$a_j = \cos \alpha e_1 + \sin \alpha \omega^j e_2, \quad b_j = \cos \alpha e_1 + \sin \alpha e^{i\phi} \omega^j e_2, \quad j = 0, \dots, k-1.$$

Then the $2k$ rays spanned by the a_j and b_j are pairwise distinct, and

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k(\cos^2 \alpha P_{e_1} + \sin^2 \alpha P_{e_2}).$$

Hence $(a_0, \dots, a_{k-1})$ and $(b_0, \dots, b_{k-1})$ form a pseudocontext of size k . If $\alpha \neq \pi/4$, then the pseudocontext is genuine.

Proof. Write $c = \cos \alpha$ and $s = \sin \alpha$. For each j ,

$$P_{a_j} = c^2 P_{e_1} + s^2 P_{e_2} + cs(\omega^{-j}|e_1\rangle\langle e_2| + \omega^j|e_2\rangle\langle e_1|),$$

and

$$P_{b_j} = c^2 P_{e_1} + s^2 P_{e_2} + cs(e^{-i\phi}\omega^{-j}|e_1\rangle\langle e_2| + e^{i\phi}\omega^j|e_2\rangle\langle e_1|).$$

Summing over $j = 0, \dots, k-1$ and using $\sum_{j=0}^{k-1} \omega^j = 0$ gives

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k(c^2 P_{e_1} + s^2 P_{e_2}).$$

Distinctness. Suppose $a_j = \lambda a_\ell$ for some scalar λ . Since $c \neq 0$, comparing first coordinates gives $\lambda = 1$, and then $\omega^j = \omega^\ell$, so $j = \ell$ (among $\{0, \dots, k-1\}$). Hence the a_j are pairwise distinct rays. The identical argument shows the b_j are pairwise distinct rays. If $a_j = \lambda b_\ell$, then $\lambda = 1$ and $\omega^j = e^{i\phi}\omega^\ell$, i.e. $e^{i\phi} = \omega^{j-\ell}$, which contradicts the assumption that $e^{i\phi}$ is not a k th root of unity. Hence no a_j coincides with any b_ℓ .

Genuineness. For $j \neq \ell$ within the a -tuple,

$$\langle a_j, a_\ell \rangle = c^2 + s^2 \omega^{\ell-j}.$$

This vanishes only if $c^2 = s^2$ (i.e. $\alpha = \pi/4$) and $\omega^{\ell-j} = -1$. So for $\alpha \neq \pi/4$ no two distinct vectors in the a -tuple are orthogonal. An identical computation gives $\langle b_j, b_\ell \rangle = c^2 + s^2 \omega^{\ell-j}$, so the same conclusion holds for the b -tuple. Hence neither tuple is a context when $\alpha \neq \pi/4$. $\square$

Remark 10. At the special value $\alpha = \pi/4$, the structure operator becomes $\frac{k}{2} I_{\text{span}\{e_1, e_2\}}$, so the construction yields a unit-norm tight frame on the two-dimensional support. For $k = 2$ this reduces to a context; for $k \geq 3$ it gives a highly symmetric pseudocontext (cf. Remark 12).

6 A symmetric $k = 3$ example

The case $k = 3$ admits a very transparent planar representation; it is the finite-dimensional analogue of the Mercedes-Benz frame.

Example 11 (Mercedes–Benz type pseudocontext). Fix a two-dimensional subspace $V \subset H$ with orthonormal basis (e_1, e_2) , and fix an angle $\theta \in (0, \pi) \setminus \{0, \pi/3, 2\pi/3, \pi\}$ (modulo π). Put, for $j = 0, 1, 2$,

$$a_j = \cos\left(\frac{2\pi j}{3}\right)e_1 + \sin\left(\frac{2\pi j}{3}\right)e_2, \quad b_j = \cos\left(\frac{2\pi j}{3} + \theta\right)e_1 + \sin\left(\frac{2\pi j}{3} + \theta\right)e_2.$$

Then the six rays are pairwise distinct, and

$$\sum_{j=0}^2 P_{a_j} = \sum_{j=0}^2 P_{b_j} = \frac{3}{2}I_V.$$

Thus $\{a_0, a_1, a_2\}$ and $\{b_0, b_1, b_2\}$ form a genuine pseudocontext of size $k = 3$ over $\mathbb{R}$.

Proof. In the basis (e_1, e_2) , each projector has the form

$$P_{\cos \varphi e_1 + \sin \varphi e_2} = \begin{pmatrix} \cos^2 \varphi & \cos \varphi \sin \varphi \\ \cos \varphi \sin \varphi & \sin^2 \varphi \end{pmatrix}.$$

For $\varphi_j = 2\pi j/3$ and for $\varphi_j = \theta + 2\pi j/3$, the trigonometric identities

$$\sum_{j=0}^2 \cos(2\varphi_j) = 0, \quad \sum_{j=0}^2 \sin(2\varphi_j) = 0$$

imply that the off-diagonal terms cancel while the diagonal terms each sum to $3/2$. Hence both sums equal $\frac{3}{2}I_V$.

The exclusion $\theta \notin \{0, \pi/3, 2\pi/3, \pi\}$ (modulo π) is needed to guarantee that no vector b_j coincides with any a_ℓ or with $-a_\ell$ (same ray). Specifically: $\theta = 0$ gives $b_j = a_j$; $\theta = \pi$ gives $b_j = -a_j$ (same ray); $\theta = \pi/3$ maps the b -triple onto the a -triple (as a set); and $\theta = 2\pi/3$ similarly identifies the two sets with a permutation.

Since $c^2 \neq s^2$ at $\alpha = \pi/4$ is not applicable here (all vectors have unit norm in $\mathbb{R}^2$ with $c = \cos(2\pi j/3)$ varying), neither triple consists of mutually orthogonal vectors for generic θ , so the pseudocontext is genuine. $\square$

Remark 12 (Complex trine description). Example 11 has a compact complex analogue. Let $\omega = e^{2\pi i/3}$ and embed V into $\mathbb{C}^n$ as $\text{span}\{e_1, e_2\}$. The *trines*

$$u_j = \frac{1}{\sqrt{2}}(1, \omega^j, 0, \dots, 0), \quad v_j = \frac{1}{\sqrt{2}}(1, e^{i\phi}\omega^j, 0, \dots, 0), \quad j = 0, 1, 2,$$

yield the same phenomenon after a change of basis in V . The identity $1 + \omega + \omega^2 = 0$ makes the off-diagonal entries of $\sum_j P_{u_j}$ cancel, giving the common structure operator $\frac{3}{2}I_V$. This form is most natural in frame-theoretic discussions and in connection with the Weyl–Heisenberg group.

Remark 13 (Nondegenerate structure operators). The trine family is highly symmetric: its structure operator is a scalar multiple of the identity on its support. More complicated examples with nondegenerate structure operators occur already in dimension three; see, for instance, the coordinatisations discussed in [4, 5]. Such examples are useful when one wants to move beyond purely symmetric frame orbits.

7 Spectral structure and canonical forms

We analyse the general properties of the structure operator

$$T = \sum_{i=1}^k P_{a_i}$$

associated with a k -tuple of unit vectors $(a_1, \dots, a_k)$ in H .

7.1 Gram representation and spectral equivalence

Definition 14 (Analysis operator and Gram matrix). Let $A: \mathbb{C}^k \rightarrow H$ be defined by $Ae_i = a_i$, where (e_i) is the standard basis of $\mathbb{C}^k$. The *analysis operator* is $A^\dagger: H \rightarrow \mathbb{C}^k$, and the *Gram matrix* is

$$G := A^\dagger A = (\langle a_i, a_j \rangle)_{i,j=1}^k.$$

Proposition 15 (Frame operator factorisation). *The structure operator admits the factorisation $T = AA^\dagger$, and $G = A^\dagger A$.*

Proposition 16 (Spectral equivalence). *The nonzero spectra of T and G coincide (including multiplicities):*

$$\sigma(T) \setminus \{0\} = \sigma(G) \setminus \{0\}.$$

In particular,

$$\text{rank}(T) = \text{rank}(G) \leq \min(k, n).$$

Proof. This is the standard singular-value correspondence between $AA^\dagger$ and $A^\dagger A$: if $AA^\dagger v = \lambda v$ with $\lambda \neq 0$, then $A^\dagger A(A^\dagger v) = \lambda(A^\dagger v)$ and $A^\dagger v \neq 0$. $\square$

7.2 Spectral constraints

Proposition 17 (Trace identities). *The structure operator satisfies*

$$\text{tr}(T) = k, \quad \text{tr}(T^2) = \sum_{i,j=1}^k |\langle a_i, a_j \rangle|^2.$$

The diagonal terms $i = j$ each contribute 1, giving $\text{tr}(T^2) \geq k$, with equality if and only if the vectors are mutually orthogonal.

Proof. $\text{tr}(T) = \sum_i \text{tr}(P_{a_i}) = k$. For the second identity, $\text{tr}(T^2) = \sum_{i,j} \text{tr}(P_{a_i} P_{a_j}) = \sum_{i,j} |\langle a_i, a_j \rangle|^2$. $\square$

Proposition 18 (Non-orthogonality and spectral spread). *If the vectors (a_i) are not mutually orthogonal, then*

$$\text{tr}(T^2) > k.$$

Equivalently, the spectrum of T is non-flat and $\lambda_{\max} > 1$.

Proposition 19 (Majorisation constraint). *Suppose $k \leq n$. Let $\lambda(T) = (\lambda_1, \dots, \lambda_n)$ be the eigenvalues of T in decreasing order (padded with zeros), and let $d = (1, \dots, 1, 0, \dots, 0)$ be the vector with k ones and $n - k$ zeros. Then*

$$\lambda(T) \succ d.$$

Proof. The diagonal entries of the Gram matrix G are $G_{ii} = \|a_i\|^2 = 1$. By the Schur–Horn theorem, the vector of diagonal entries of G is majorised by the vector of its eigenvalues. Since G and T share the same nonzero eigenvalues (Proposition 16), the eigenvalue vector of T (padded with $n - k$ zeros when $k \leq n$) majorises $d = (1, \dots, 1, 0, \dots, 0)$.

Note on the overcomplete case $k > n$. When $k > n$ the Gram matrix G is $k \times k$ with rank at most n ; the majorisation statement must then be formulated for the k -dimensional eigenvalue vector of G against $(1, \dots, 1) \in \mathbb{R}^k$, and the above argument applies to G in place of T . $\square$

Proposition 20 (Operator bounds). *The structure operator satisfies*

$$0 \leq T \leq kI.$$

Moreover, if $\mu := \max_{i \neq j} |\langle a_i, a_j \rangle|$, then

$$\lambda_{\max}(T) \leq 1 + (k - 1)\mu.$$

Proof. Positivity $T \geq 0$ is immediate, and $T \leq kI$ since each $P_{a_i} \leq I$. For the sharper bound, apply the Geršgorin circle theorem to the Gram matrix G : since $G_{ii} = 1$ and $|G_{ij}| \leq \mu$ for $i \neq j$, every eigenvalue of G lies in a disk centred at 1 with radius $\sum_{j \neq i} |G_{ij}| \leq (k - 1)\mu$. Because the nonzero eigenvalues of T and G coincide, $\lambda_{\max}(T) \leq 1 + (k - 1)\mu$. $\square$

7.3 Commutant and symmetry

Definition 21 (Commutant). Let

$$\mathcal{C}(T) := \{U \in U(H) : UT = TU\}$$

be the unitary commutant of T .

Proposition 22 (Structure of the commutant). *Let $T = \sum_{\alpha} \lambda_{\alpha} P_{\alpha}$ be the spectral decomposition of T with distinct eigenvalues λ_{α} and orthogonal projections P_{α} of ranks m_{α} . Then*

$$\mathcal{C}(T) \cong \prod_{\alpha} U(m_{\alpha}).$$

Remark 23. Spectral degeneracies ($m_{\alpha} > 1$) give rise to continuous families of distinct rank-one decompositions of T (as in tight frames), whereas simple spectrum leads to strong rigidity. This explains why the tight-frame (scalar-multiple-of-identity) case in Remark 10 is so rich in pseudocontexts.

7.4 Canonical normal form of the structure operator

Proposition 24 (Spectral normal form). *Every structure operator is unitarily equivalent to a diagonal operator*

$$T \cong \text{diag}(\lambda_1, \dots, \lambda_r, 0, \dots, 0),$$

where $\lambda_i > 0$, $\sum_{i=1}^r \lambda_i = k$, and $r = \text{rank}(T)$.

7.5 Canonical parametrisation of all decompositions

We now describe all rank-one decompositions of a fixed positive operator T .

Theorem 25 (Canonical decomposition via the analysis operator). *Let T be a positive operator of rank r with spectral decomposition*

$$T = \sum_{i=1}^r \lambda_i |e_i\rangle\langle e_i|, \quad \lambda_i > 0,$$

and set $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_r)$.

Every rank-one decomposition $T = \sum_{j=1}^k P_{a_j}$ with $\|a_j\| = 1$ is obtained as follows. There exists a matrix $C \in \mathbb{C}^{r \times k}$ such that

$$CC^\dagger = \Lambda, \quad \|c_j\|^2 = 1 \text{ for all } j, \quad (5)$$

where c_j is the j -th column of C , and $a_j = \sum_{i=1}^r c_{ij} e_i$. Conversely, every matrix C satisfying (5) defines a valid decomposition.

A necessary condition for such a C to exist is that $\lambda_i \leq k - r + 1$ for all i [11].

Proof. Let $T = AA^\dagger$ be any factorisation with columns a_j . In the eigenbasis of T , write $A = UC$ where U diagonalises T (restricted to $\text{supp}(T)$). Then $T = AA^\dagger = UCC^\dagger U^\dagger$, so $CC^\dagger = \Lambda$. The column-norm condition follows from $\|a_j\| = 1$. The converse is a direct computation. $\square$

Corollary 26 (Unitary freedom). *Among all rank-one decompositions of T into k (not necessarily unit-norm) vectors, any two are related by right multiplication $A \mapsto AU$ for some $U \in U(k)$. Imposing the unit-norm constraint (5) cuts out a submanifold of this orbit; in general, the $U(k)$ -action is not transitive on the unit-norm decompositions.*

7.6 Characterisation of pseudocontexts

Proposition 27 (Operator equivalence criterion). *Two k -tuples (a_i) and (b_i) of distinct rays form a pseudocontext if and only if their corresponding matrices C_A, C_B (expressed in the eigenbasis of T) satisfy*

$$C_A C_A^\dagger = C_B C_B^\dagger = \Lambda.$$

Remark 28. Thus pseudocontexts correspond to distinct points in the space

$$\{C \in \mathbb{C}^{r \times k} : CC^\dagger = \Lambda, \|c_j\| = 1\} / \sim,$$

where $\sim$ identifies decompositions giving the same set of rays. This is the space of inequivalent rank-one decompositions of a fixed positive operator with prescribed column norms.

7.7 Extremal spectral bounds

Proposition 29 (Rayleigh bounds). *For any density operator ρ ,*

$$\lambda_{\min}(T) \leq \text{tr}(\rho T) \leq \lambda_{\max}(T).$$

Remark 30. These bounds are tight (achieved by pure states supported on the eigenvectors of T). They determine the quantum range of the probability sums $\sum_i P(a_i)$ that define the pseudocontext.

7.8 Examples of nondegenerate spectra

Explicit three-dimensional implementations of nondegenerate structure operators satisfying the bounds above were derived in [4] in the study of generalised gadget hypergraphs.

- **15-atom hypergraph (T_1).** A pseudocontext of size $k = 3$ generated from a generalised firefly logic yields a structure operator T_1 with eigenvalues

$$\lambda_1 = 2, \quad \lambda_2 = \frac{7+\sqrt{21}}{14} \approx 0.827, \quad \lambda_3 = \frac{7-\sqrt{21}}{14} \approx 0.173.$$

One checks $\text{tr}(T_1) = 3$ and $\lambda_{\max} = 2 > 1$, consistent with Proposition 18. The Geršgorin bound of Proposition 20 is satisfied as well.

- **36-atom hypergraph (T_2).** A larger generalised true-implies-false gadget gives another $k = 3$ pseudocontext. Its structure operator is $T_2 = \text{diag}(1/5, 1/5, 13/5)$. Here $\text{tr}(T_2) = 3$ and the spatial frame is stretched anisotropically with $\lambda_{\max} = 13/5 > 1$.

8 Physical interpretation and constraints

A pseudocontext $(\{a_i\}, \{b_i\})$ implements an empirical equivalence between two families of rank-one observables $A_i = P_{a_i}$ and $B_i = P_{b_i}$. Although the operators within each family are generally noncommuting, the two linear functionals

$$\rho \mapsto \sum_i \text{tr}(\rho A_i), \quad \rho \mapsto \sum_i \text{tr}(\rho B_i)$$

agree on the whole state space. Because the structure operators are identical, this equivalence is strictly state-independent.

The spectral decomposition of T establishes absolute quantum limits on these probability sums. By the Rayleigh–Ritz theorem (Proposition 29),

$$\lambda_{\min}(T) \leq \sum_{i=1}^k P(a_i) \leq \lambda_{\max}(T). \quad (6)$$

These bounds are tight. For the 36-atom hypergraph example [4], the structure operator $T_2 = \text{diag}(1/5, 1/5, 13/5)$ confines the quantum probability sum to the interval $[1/5, 13/5] = [0.2, 2.6]$. A classical hidden-variable model based on two-valued $(\{0, 1\})$ truth assignments to projections—subject to the consistency requirement that exactly one atom in each orthogonal basis receives the value 1—would allow the sum to range over the integer values $\{0, 1, 2, 3\}$, which includes 0 and 3. Because T_2 has no eigenvalue equal to 0 or 3, quantum mechanics forbids these extreme classical values; the quantum system cannot reach them. This discrepancy is a signature of Kochen–Specker-type contextuality [6].

We note that this state-independent additive equivalence is related to, but distinct from, the operational notion of contextuality introduced by Spekkens [8], which applies to arbitrary experimental procedures (preparations, transformations, and unsharp measurements) rather than to sharp projective measurements alone. A precise comparison is beyond the scope of this note.

In the language of quantum information theory, the collections $\{A_i\}$ and $\{B_i\}$ are distinct rank-one operator decompositions of the same positive operator. If the common

operator is proportional to the identity on its support, they form a tight frame; after normalisation one obtains a POVM. From a foundational standpoint, this is a state-independent additive equivalence closely related to, but distinct from, Kochen–Specker-type contextuality [6, 7].

9 Open problems

Having established the existence of pseudocontexts for every size $k \geq 2$ and isolated a simple $k = 3$ illustration, we highlight the remaining questions regarding their deeper classification.

Open Problem (Classification). Classify, up to unitary equivalence, all pseudocontexts of size k in $\mathbb{C}^n$. Even for small (k, n) , the orbit structure is not yet transparent.

Open Problem (Lattice representability). Characterise which orthomodular lattices carrying a pseudocontext-type additive identification admit a faithful embedding into some $L(\mathbb{C}^n)$, sharpening the partial results of [4, 5].

Open Problem (Frame-symmetry characterisation). Can one characterise pseudocontexts purely in terms of symmetries of the underlying rank-one frame—for example, in terms of unitaries commuting with the structure operator and permuting the atoms of the two decompositions?

Open Problem (Minimal ambient dimension). Although every pseudocontext of size $k \geq 2$ can be realised in dimension $n = 2$, what is the minimal n for which *all* pseudocontexts of a given combinatorial type occur? For Kochen–Specker configurations this question has a long history; the pseudocontext analogue is open.

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